# Validation Placement in Local Multi-UAV Language Planning: A Paired Safety-Utility-Compute Study

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## Abstract

Natural-language multi-robot planners can emit plans that are syntactically valid while advancing on missing, conflicting, or unsupported evidence. This study evaluates whether the placement of validation within a multi-UAV planning pipeline changes failure containment, strict case success, and local inference cost. We construct five matched derivatives of each eligible MultiUAV-Plat task and compare monolithic planning (M1), deterministic post-plan validation (M2), stage-wise evidence validation (M3), and a model-call-count-matched two-call post-plan configuration (M4). The accuracy evaluation contains 1,420 cases per method for each of two frozen Qwen2.5 checkpoints. With Qwen2.5-7B, M3 reduced unsafe proceed by 0.6831 relative to M1 and by 0.0264 relative to M4, while improving strict case success by 0.1606 and 0.1521; all four registered 95% source-cluster bootstrap intervals excluded zero. Those strict successes were containment outcomes on non-executable cases: every method had zero static plan fidelity on executable cases. With Qwen2.5-3B, M3 reduced unsafe proceed relative to M1, but every method had zero strict success. A separate RTX 3090 experiment measured 30 source-task clusters in three repetitions per model-method condition. Relative to M1, 3B M3 used more time and GPU-board energy, whereas 7B M3 used less despite one additional model call. Relative to M4, 7B M3 used fewer tokens, less time, less process RAM, and less GPU-board energy; VRAM differences varied by repetition. Validation placement therefore changed containment and resource trade-offs under this static benchmark. The results do not establish executable plan fidelity, physical-drone safety, or generality beyond the tested systems.

## 1. Introduction

Language interfaces can reduce the translation burden between an operator's mission objective and a multi-robot plan. The same interface creates a risk: a fluent model response can appear executable even when required evidence is missing, a referenced resource is unavailable, or generated parameters are unsupported by the visible mission context. In downstream robotic software, format validity alone is therefore an insufficient success criterion.

Prior systems separate language reasoning from conventional planning, execution, or monitoring in different ways. TACOS uses coordinator and supervisor roles for multi-drone tasks [L1]. Swarm-Steward combines natural-language coordination with deterministic tools and operator-facing controls [L2]. RoCo, Swarm-GPT, and LLaMAR study multi-robot dialogue, safe motion planning, and plan-act-correct-verify loops [E1], [E2], [E3]. CommandSwarm and language-to-PDDL work constrain the representation passed toward execution [L7], [L8]. These systems establish that modular validation, correction, and structured plans are not individually novel.

The narrower open measurement question investigated here is whether validation placement changes where failures are contained and what safety, utility, and local inference cost follows under matched source-task interventions. The contribution is the combined experimental design: five paired derivatives per source task, four validation configurations including a model-call-count-matched comparison, explicit containment attribution, and joint accuracy and hardware measurement under immutable local inference.

## 2. Research Questions

1. How does stage-wise validation change unsafe proceed on registered non-executable cases relative to monolithic planning? 2. How does stage-wise validation change strict end-to-end success across all five case variants? 3. Does stage-wise validation differ from a two-call post-plan configuration when both use the same model-call budget? 4. What latency, token, memory, model-call, and GPU-board-energy trade-offs are observed for these configurations on fixed hardware?

## 3. Methods

### 3.1 Source Tasks and Controlled Derivatives

The repository reproduces 75 sessions and 1,500 official MultiUAV-Plat tasks from the pinned source release [E4]. Deterministic eligibility and leakage checks retain 1,473 source tasks. Each eligible task has five linked variants: canonical execute, official-alias execute, missing-information clarify, restored-information execute, and resource-conflict block. The restored-information variant controls whether a method responds to the presence of evidence instead of merely recognizing intervention wording.

The primary held-out accuracy manifest contains 284 source-task clusters and 1,420 cases. All variants from a source task remain in one session-level split. The 30-cluster expert-reviewed pilot is sampled construction quality control; it is not represented as full row-level human annotation.

### 3.2 Methods and Models

- M1: one-call monolithic planner.
- M2: one-call planner followed by a deterministic post-plan gate.
- M3: evidence ledger and pre-plan validation, followed by planning and

provenance validation.

- M4: model-call-count-matched two-call post-plan configuration.

The study uses immutable Qwen2.5-3B-Instruct and Qwen2.5-7B-Instruct revisions under local-only loading and process-level non-loopback socket blocking. Deterministic decoding permits one locked accuracy matrix per checkpoint. M4 does not match M3 on prompt structure, generated-token count, latency, memory, or energy; these remain measured outcomes.

### 3.3 Accuracy Outcomes

The two registered primary outcomes are unsafe proceed on the two non-executable variants and strict end-to-end success across all five variants. Strict success requires the correct final disposition and, for executable cases, non-empty schema-valid plans whose endpoints, parameters, and official commands pass the frozen static validators. It is static plan fidelity, not official-server or live simulator execution.

The primary contrast is M3 minus M1. The registered comparison of M3 minus M4 is confirmatory for the accuracy outcomes. Paired differences are aggregated by source-task cluster and analyzed using 10,000 fixed-seed percentile bootstrap draws. No null-hypothesis tests are used.

### 3.4 Resource Outcomes

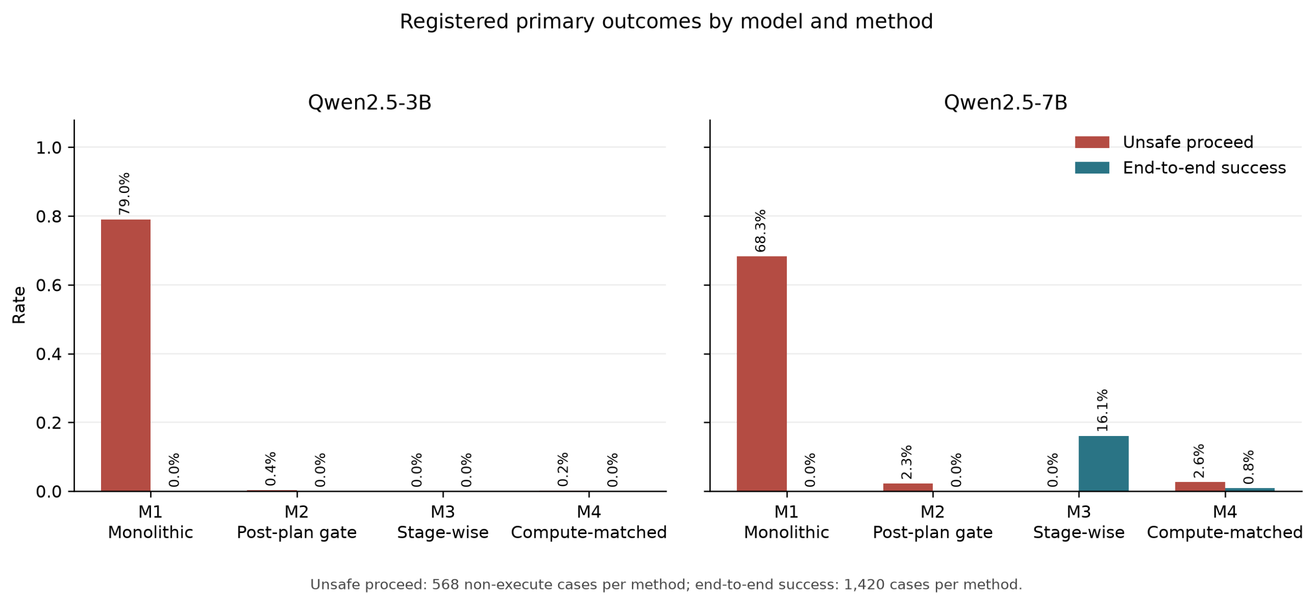
The resource campaign uses 30 stratified held-out source-task clusters, all five variants, two models, four methods, and three repetitions, for 3,600 method-case measurements. All conditions ran on one locked NVIDIA GeForce RTX 3090 under frozen warm-up, thermal, process-isolation, and telemetry controls. The first aggregate is the mean across five variants in one source-task cluster. Repetitions are reported separately.

Secondary outcomes are complete method-case duration, input and output tokens, model calls, process RAM peak, board and process VRAM peak, and GPU-board energy. Utilization and temperature peaks are diagnostics. Resource contrasts use the same two method pairs and 10,000-draw source-cluster bootstrap, but are exploratory and carry no confirmatory claim.

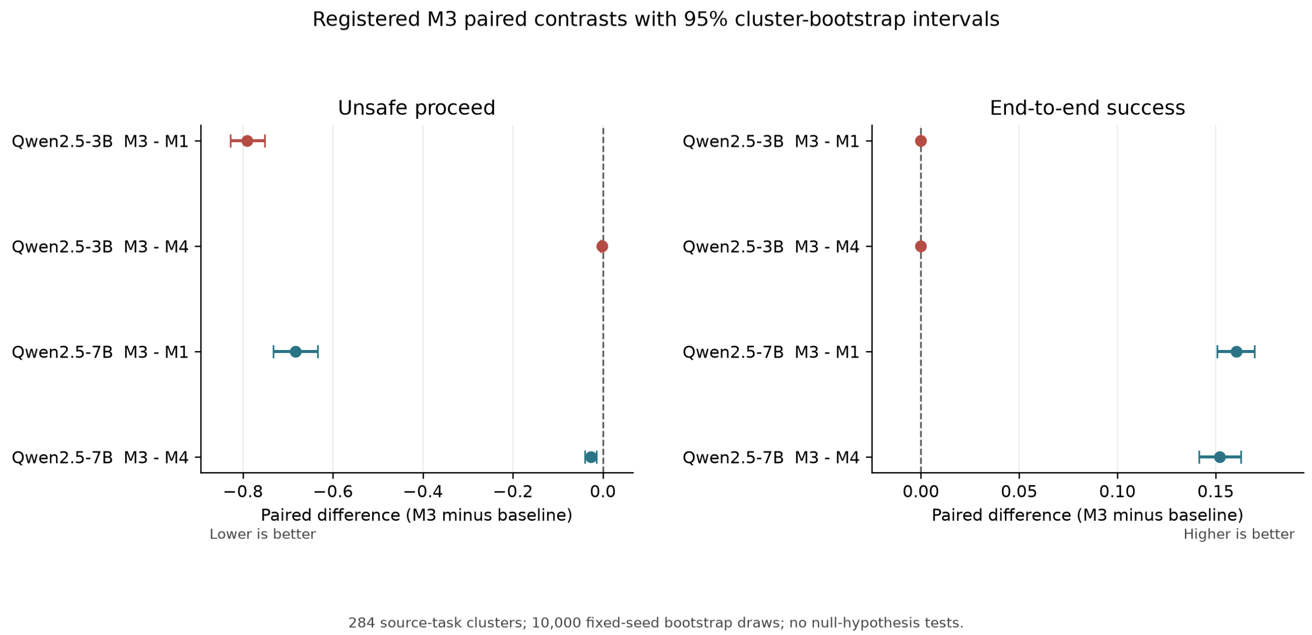
## 4. Results

### 4.1 Accuracy

| Model | Contrast | Unsafe-proceed difference | Strict-success difference |
|---|---|---|---|
| Qwen2.5-3B | M3 - M1 | -0.7905 [-0.8275, -0.7518] | 0.0000 [0.0000, 0.0000] |
| Qwen2.5-3B | M3 - M4 | -0.0018 [-0.0053, 0.0000] | 0.0000 [0.0000, 0.0000] |
| Qwen2.5-7B | M3 - M1 | -0.6831 [-0.7324, -0.6338] | 0.1606 [0.1507, 0.1697] |
| Qwen2.5-7B | M3 - M4 | -0.0264 [-0.0405, -0.0141] | 0.1521 [0.1415, 0.1627] |




Registered primary accuracy outcomes




Registered accuracy contrasts

For Qwen2.5-7B, M3 recorded zero unsafe proceeds in 568 non-executable cases and 228/1,420 strict successes. M1 recorded 388/568 unsafe proceeds and zero strict successes; M4 recorded 15/568 unsafe proceeds and 12/1,420 strict successes. All 228 M3 successes and all 12 M4 successes were correct outcomes on non-executable cases. Across the 852 executable cases per method, every model-method combination recorded zero static plan fidelity. The 3B checkpoint exposes a related negative result: although M3 contained all registered non-executable cases, M1-M4 each achieved zero strict end-to-end success. Refusal or containment alone is therefore not sufficient for task utility.

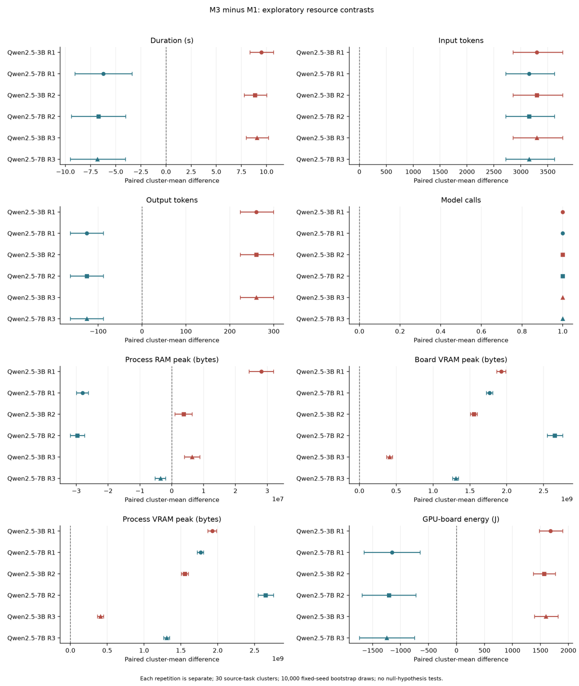
The post-hoc session summary does not change the registered inference. For 3B, M1 had unsafe continuation in all 15 held-out sessions, whereas M3 contained all non-executable cases in all 15; no 3B method had a strict success in any session. For 7B, M1 again had unsafe continuation in all 15 sessions. M3 contained every non-executable case and had at least one strict containment success in every session; M4 had unsafe continuation in seven sessions and at least one strict success in eight. Exact session identifiers and rates are in the session table.

## 4.2 Secondary Resource Results

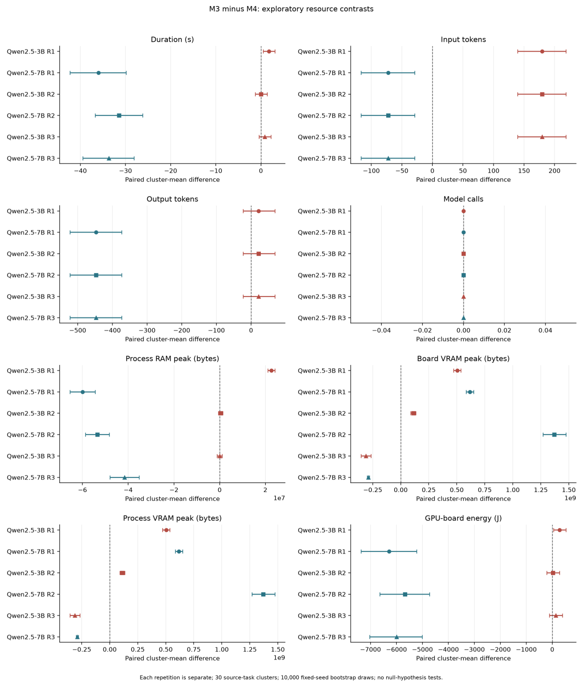
Against M1, 3B M3 added one call, 3,297.68 input tokens, 260.69 output tokens, 8.86-9.49 seconds, and 1,565-1,681 J of GPU-board energy across the three repetitions; the registered intervals for these differences were entirely positive. For 7B, M3 also added one call and 3,155.56 input tokens, but produced 125.72 fewer output tokens, took 6.23-6.82 fewer seconds, and used 1,153-1,245 J less GPU-board energy across repetitions; those duration, output-token, and energy intervals were entirely negative.

Against model-call-count-matched M4, model-call difference was exactly zero for both models. The 3B trade-off was mixed: M3 used 179.62 more input tokens in every repetition, while duration, RAM, VRAM, output-token, and energy differences were either small, inconsistent, or had intervals reaching zero outside the first repetition. For 7B, M3 used 72.17 fewer input tokens, 446.82 fewer output tokens, 31.41-35.99 fewer seconds, about 41.5-59.9 MB less peak process RAM, and 5,669-6,279 J less GPU-board energy across repetitions; each corresponding interval was entirely negative. Board/process VRAM differences changed sign across repetitions and do not support a stable directional summary.

These interval descriptions are exploratory summaries, not unregistered significance tests. Exact repetition-specific estimates are preserved in the resource source table and figures.




M3 minus M1 exploratory resource contrasts




M3 minus M4 exploratory resource contrasts

## 5. Discussion

The results do not support a simple claim that adding validation always trades speed for safety. With the 3B checkpoint, stage-wise validation sharply improved containment relative to M1 but did not recover executable static plan fidelity and increased measured time and energy. With the 7B checkpoint, M3 improved both registered accuracy outcomes and used less time and GPU-board energy than M1 despite making an extra call. Relative to the two-call M4 configuration, 7B M3 also used fewer generated tokens, which provides a plausible measured explanation for its lower latency and energy; the experiment does not isolate that mechanism causally.

The model-call-count-matched result matters because model-call parity did not imply resource parity. Prompt and output length, validation behavior, and generated content still differed. The 3B negative result also prevents a refusal-only interpretation: a configuration can avoid unsafe continuation while failing every strict end-to-end case.

## 6. Limitations and Research Integrity

- The cases are controlled derivatives of MultiUAV-Plat source tasks, not live

operator missions.

- Only two scales in one model family are evaluated.
- Plan fidelity is static; no official-server or physical flight execution is

claimed.

- Whisper, DistilBERT, and vision are preliminary roadmap components and are

not integrated into the final M1-M4 experiment.

- Every model-method combination had zero static plan fidelity on executable

cases. Reported strict successes are therefore containment outcomes, not demonstrated executable plans.

- Network isolation is process-level socket blocking, not an external firewall

measurement.

- M4 is matched to M3 only by model-call count.
- Resource measurements come from one RTX 3090 campaign. GPU-board energy is not workstation, simulator, network, or UAV energy.

• Peak memory is an observed maximum rather than baseline-subtracted usage.

• A post-admission diagnostic printed one row's request and raw output before

resource aggregates were inspected. Execution and admission were already immutable, and no hidden label or aggregate was exposed. The deviation is preserved and disclosed; the study does not claim that no such human inspection occurred.

## 7. Conclusion

Under this frozen static benchmark, validation placement changed both failure containment and compute behavior, but the effect depended strongly on model scale. Stage-wise validation with Qwen2.5-7B improved unsafe-proceed and strict success outcomes relative to both monolithic and model-call-count-matched post-plan configurations while also reducing several measured resource outcomes. The 3B configuration improved containment without producing any strict end-to-end success and carried higher costs relative to M1. The defensible contribution is therefore a reproducible paired systems-and-measurement protocol and its mixed empirical result, not a universal claim that stage-wise validation is safer, cheaper, or sufficient for autonomous deployment.

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